

## PROGRAM 2:

### Main.java

```
package com.example;
public class App {
    public int add(int a, int b) {
        return a + b;
    }
    public static void main(String[] args) {
        App app = new App();
        int result = app.add(2, 3);
        System.out.println("2 + 3 = " + result);
        System.out.println("Application executed successfully!");
    }
}
```

### Test.java

```
package com.example;
import org.junit.Assert;
import org.junit.Test;
public class AppTest {
    @Test
    public void testAdd() {
        App app = new App();
        int result = app.add(2, 3);
        System.out.println("Running test: 2 + 3 = " + result);
        Assert.assertEquals(5, result);
    }
}
```

## Program 3

```
plugins {  
    id 'application'  
}  
application {  
    mainClass = 'com.example.AdditionOperation'  
}  
repositories {  
    mavenCentral()  
}  
dependencies {  
    testImplementation 'junit:junit:4.13.2'  
}  
test {  
    outputs.upToDateWhen { false }  
    testLogging {  
        events "passed", "failed", "skipped"  
        exceptionFormat "full"  
        showStandardStreams = true  
    }  
}
```

### **Addition.java**

```
package com.example;  
  
public class AdditionOperation {  
    public static void main(String[] args) {  
        double sum = 5 + 10;  
        System.out.println("Sum: " + sum);  
    }  
}
```

### TEST.JAVA

```
package com.example;  
  
import org.junit.Test;  
import static org.junit.Assert.*;
```

```
public class AdditionOperationTest {  
    @Test  
    public void testAddition() {  
        assertEquals(15, 5 + 10, 0.01);  
    }  
}
```

## PROGRAM 4

```
plugins {  
    id 'java'  
}  
group = 'com.example'  
version = '1.0-SNAPSHOT'  
repositories {  
    mavenCentral()  
}  
dependencies {  
    testImplementation 'junit:junit:4.12'  
}  
task run(type: JavaExec) {  
    main = 'com.example.App'  
    classpath = sourceSets.main.runtimeClasspath  
}
```